

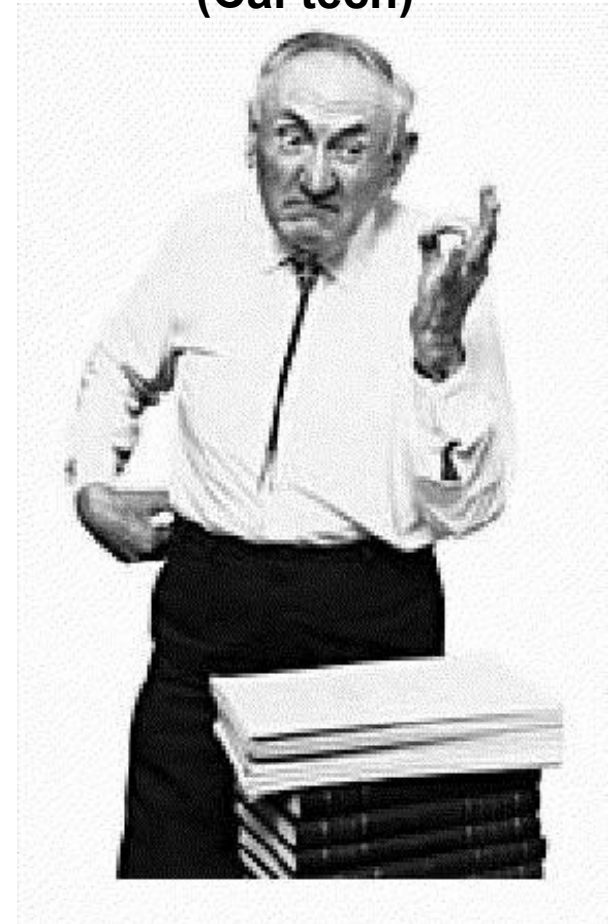
DEATH/LIFE MASSIVE STARS

- supernova type II
- neutron stars, pulsars
- black holes



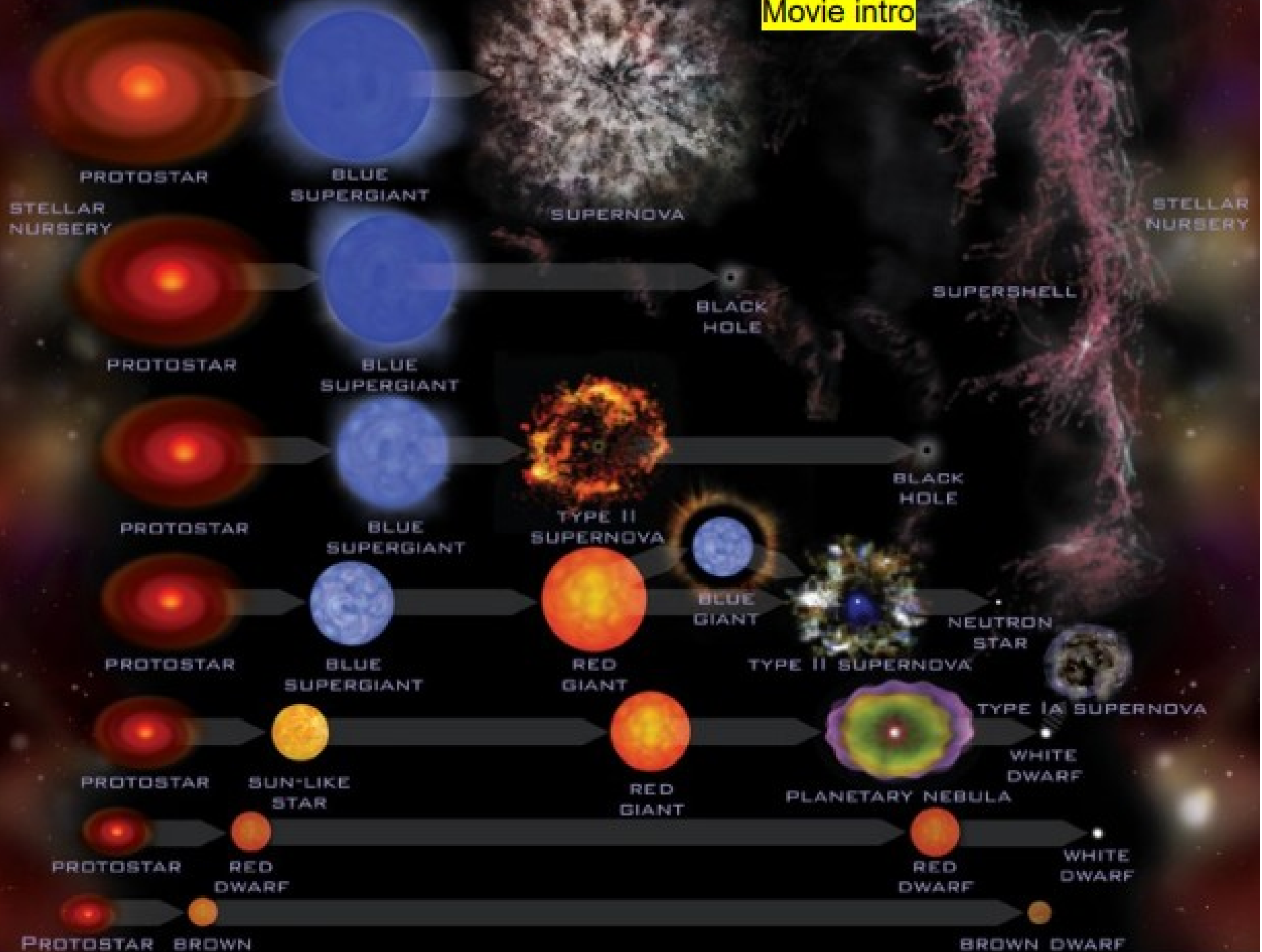
Susan Jocelyn Bell Burnell

Fritz Zwicky
(Cal tech)



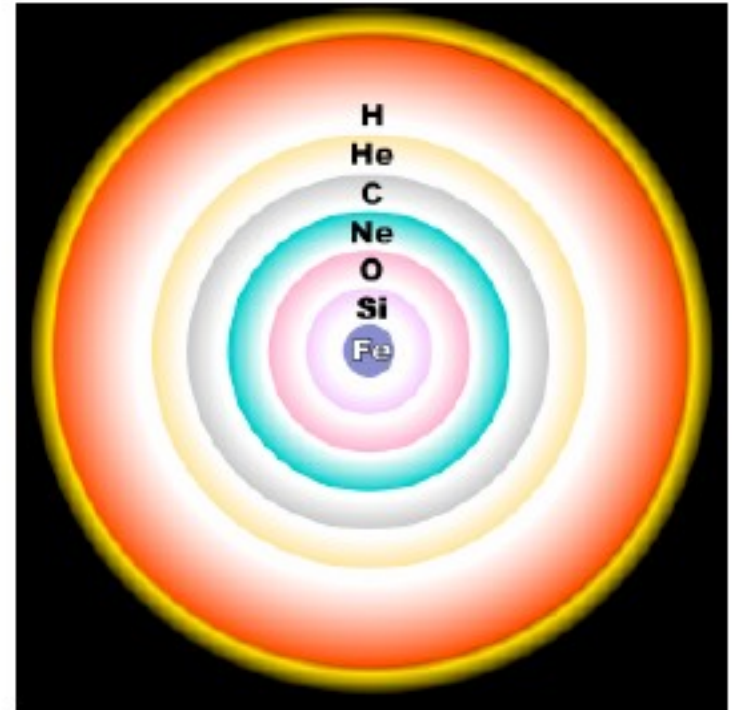
GRAVEYARD -

Movie intro



Type II Supernova: massive star

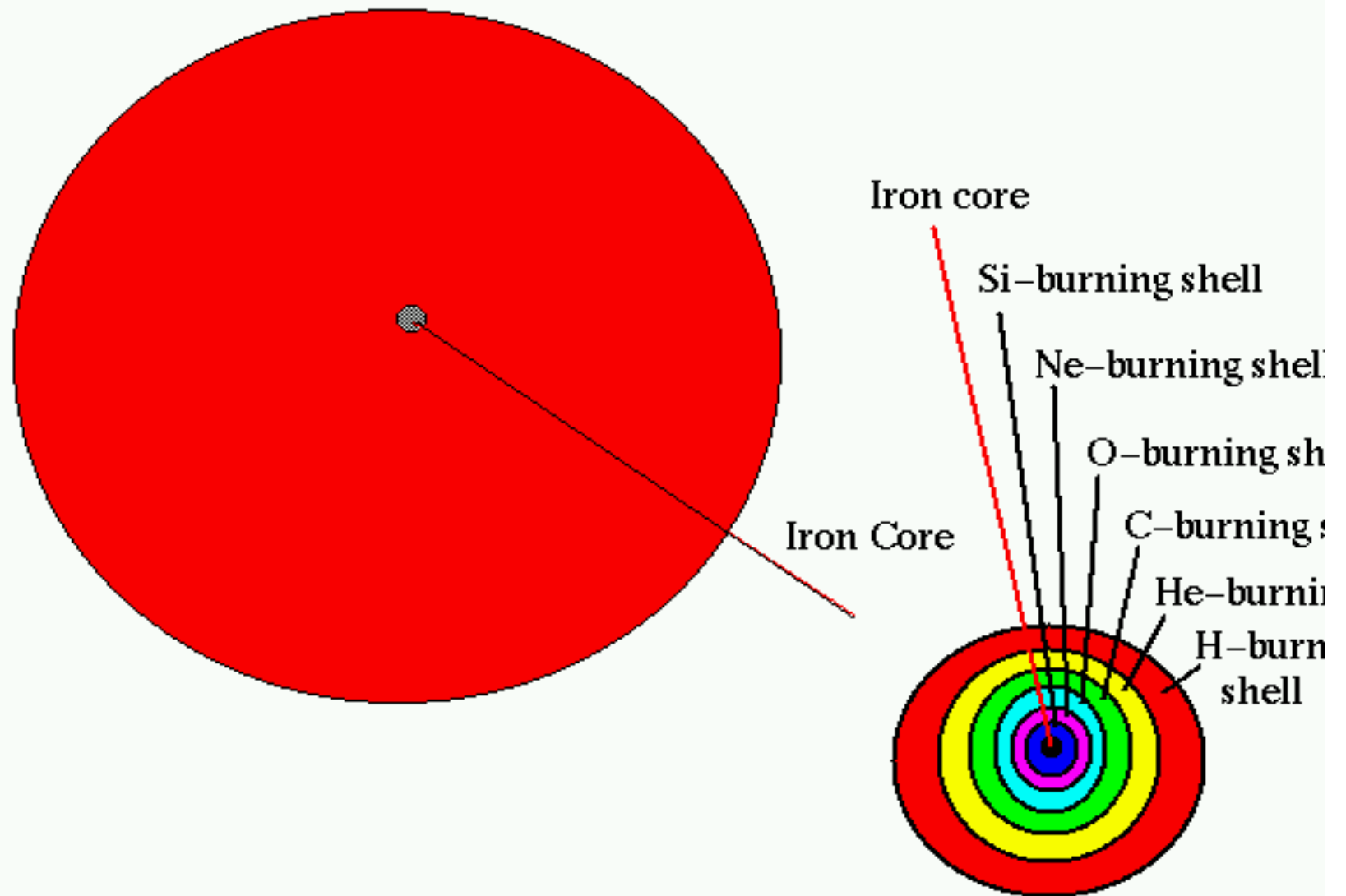
- Massive stars ($> 10 M_{\text{sun}}$) continue to burn fuel until iron (Fe) forms in the core. Fe is the **most bound** atomic nuclei. No more reactions in the Fe.
- The Fe core continues to grow.
- When the Fe core reaches $1.4M_{\text{sun}}$ the core collapses.



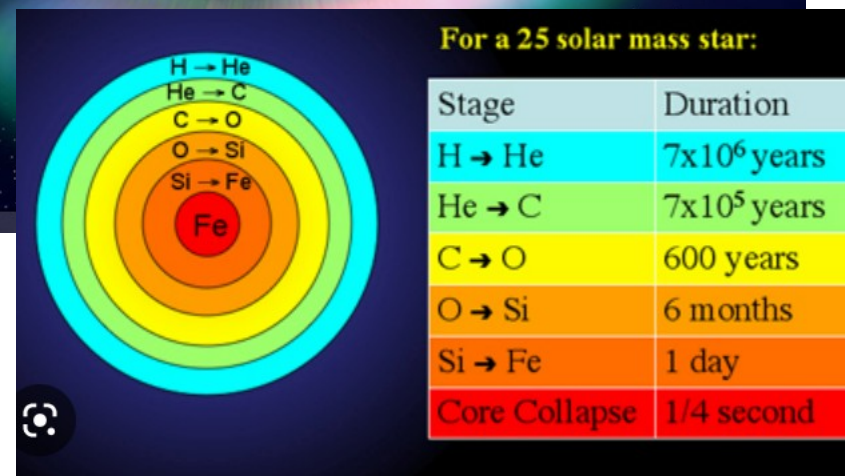
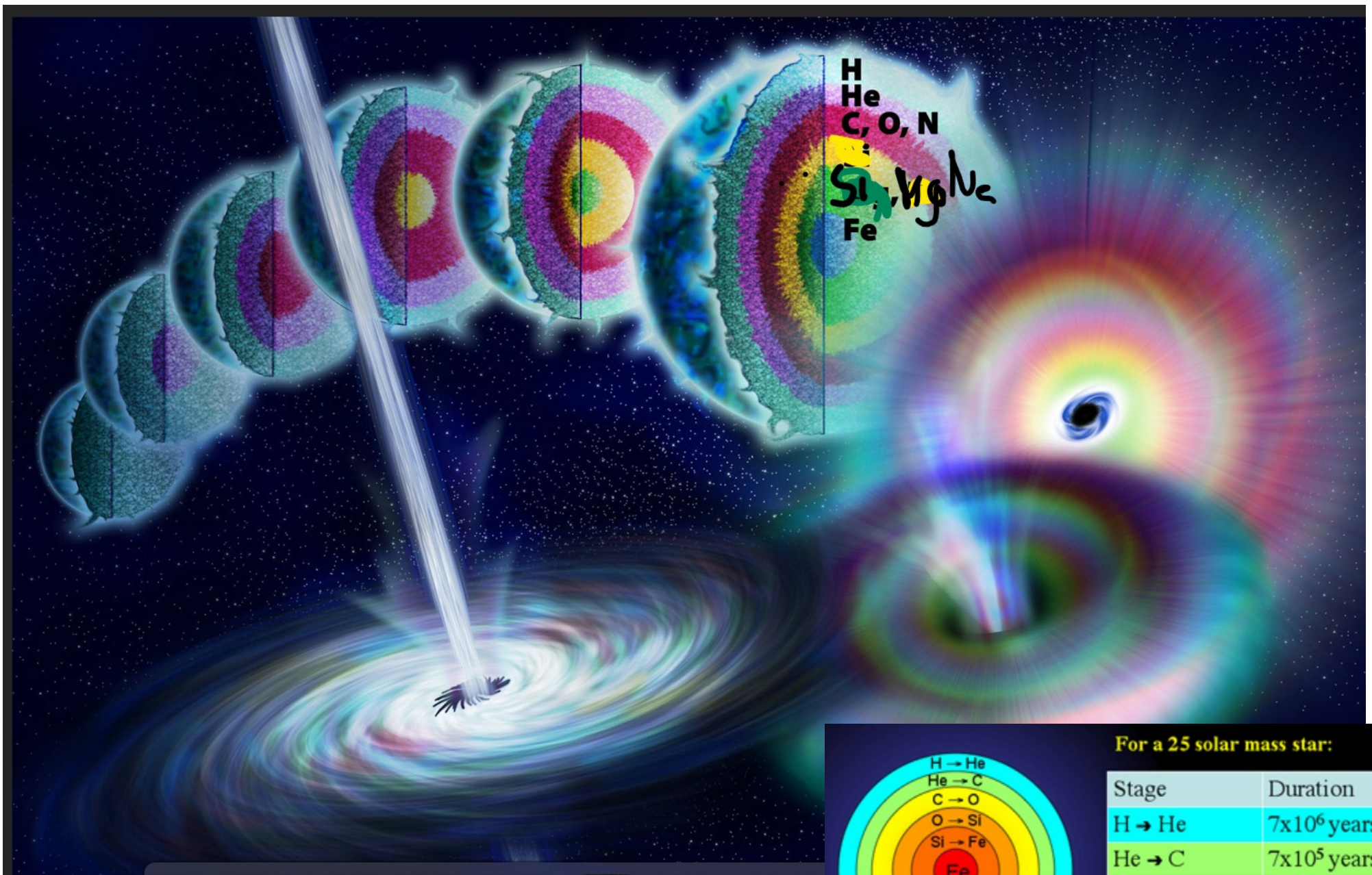
$p + e^- \rightarrow n + \nu$ The core is converted into neutrons

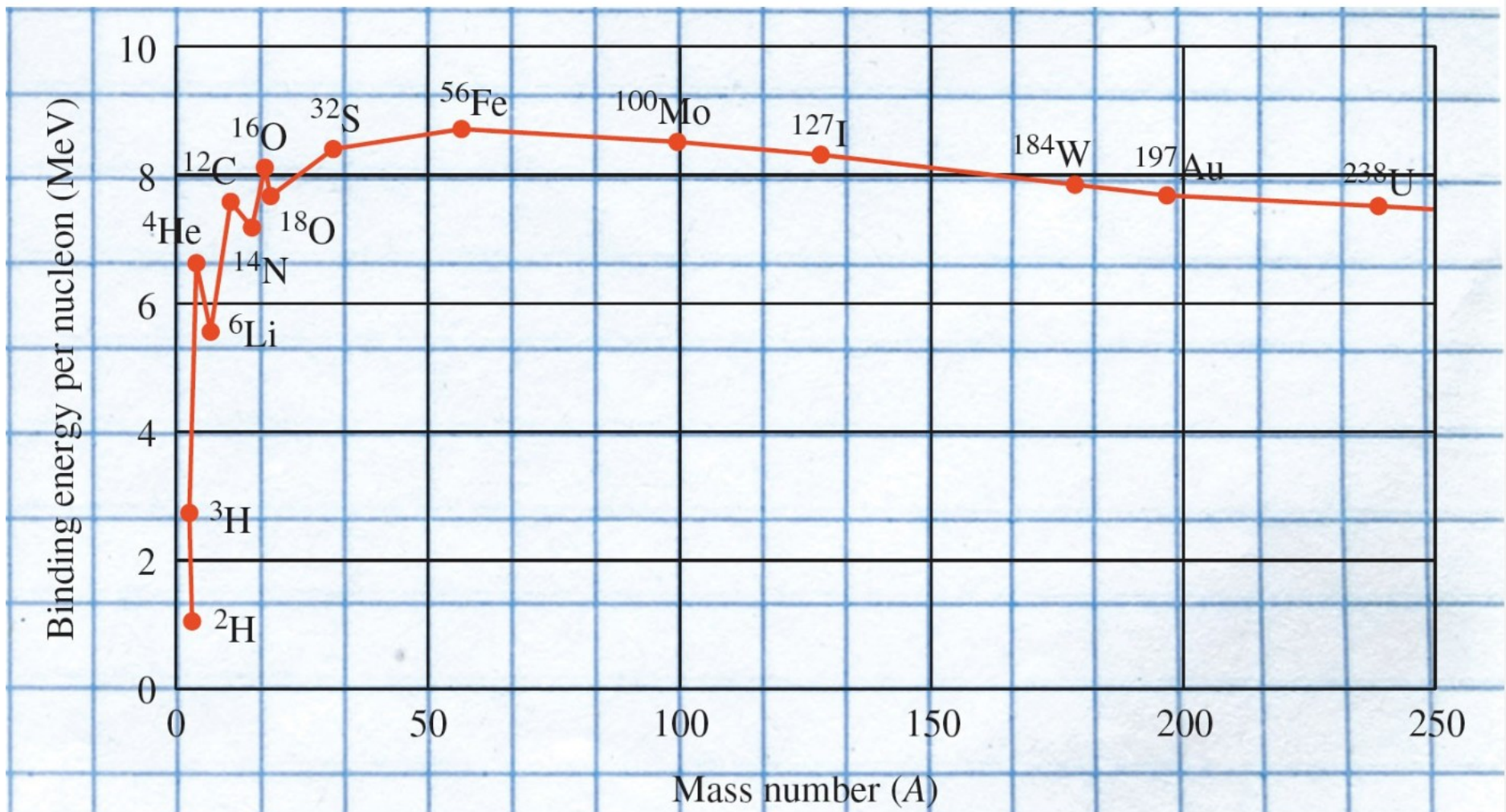
- The outer layers bounce off the core creating an explosion! (neutrinos also very important for driving the envelope away)
- A **neutron star** is formed.

Video big stars



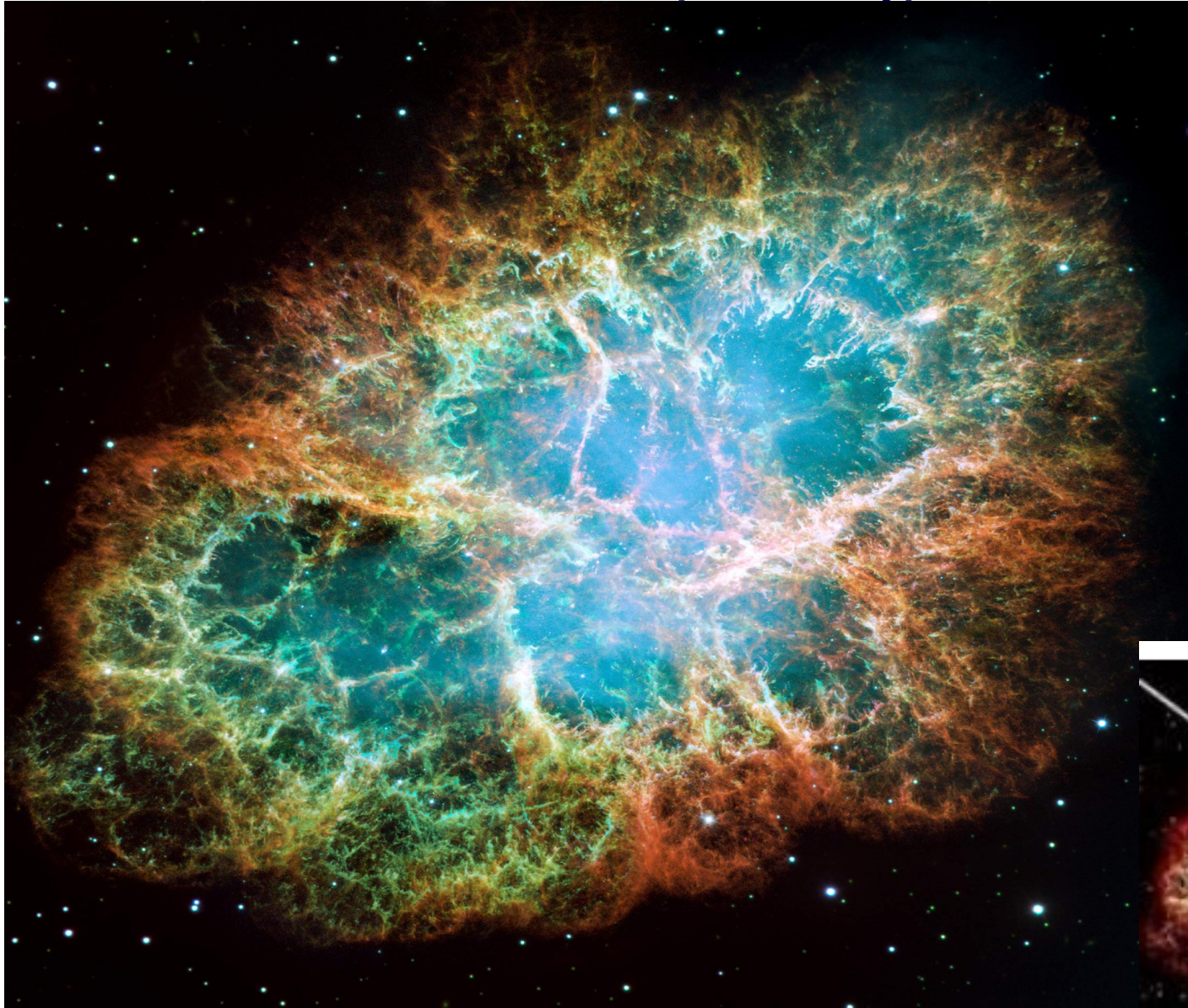
Supergiant ready to blow out





FUSION or FISSION are nuclear reactions. Both produce energy to increase the binding energy between the nucleons. The binding energy is negative. The final atoms (after fission or fusion) have less mass than the initial ones. This loss of mass was converted to energy (photons or kinetic energy). Iron is the most stable of the elements, below Iron, elements are willing to fuse. And above Iron, they are willing to split.

**Crab nebula : remnant of a supernova seen by the Chinese in 1054
Well described by them. Nothing by European (Dark age).
There is a neutron star inside. Supernovae type II**





Chaco Canyon Supernova Anazazi indian

The supernova pictograph at Chaco Culture National Historical Park.
Located in the back country below the Penasco Blanco pueblo ruin



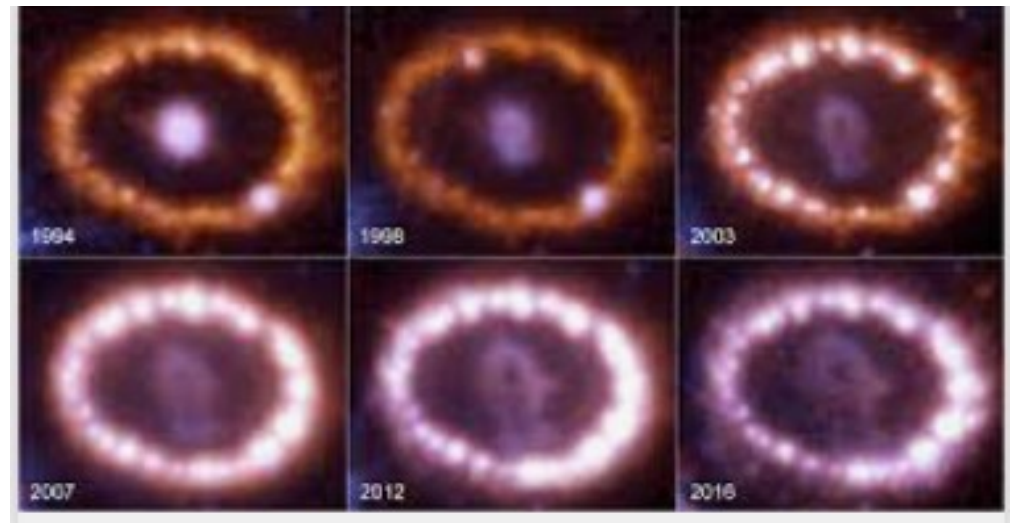
TYPE II supernovae can occur in arms of spiral galaxies where young stars are found. So they must be dying massive stars. They were born there and die there. They don't have time to migrate closer to center from their birth places. Type IA : has to do with old stars. White dwarf dying in binary system,

Supernova 1987A (Type II)

- Nearby! Only 170,000 light years away in the Large Magellanic Cloud (a small “satellite” galaxy of the Milky Way).
- It initially was a $20 M_{\text{sun}}$ star (but **blue** supergiant, NOT red -- this is a mystery, is it because of the low metallicity? Perhaps it was two stars merging?)
- Neutrinos detected! (25 of them) Explosion mechanism was core collapse and rebound. A neutron star was probably formed, but we still haven't seen it -- this is another mystery.
- This supernova confirmed many of our ideas about how stars exploded, but also brought up many new questions.

Supernova 1987A - energy output

- Total energy (emitted in about 1 second) was comparable to the energy emitted in 1 second by ALL normal stars in the entire observable Universe!
 - > 99% of the energy was in neutrinos
 - < 1% was energy of motion of ejecta
 - < 0.01% was in visible light
- Supernovae are truly incredible explosions!
- Gamma-rays with certain specific energies were seen coming from SN 1987A. This confirms that radioactive $^{56}_{27}\text{Co}$ (Cobalt) was produced.
- Confirms that **heavy elements** are made in stars and explosion and then dispersed into the Universe to form new stars, planets, and eventually life.



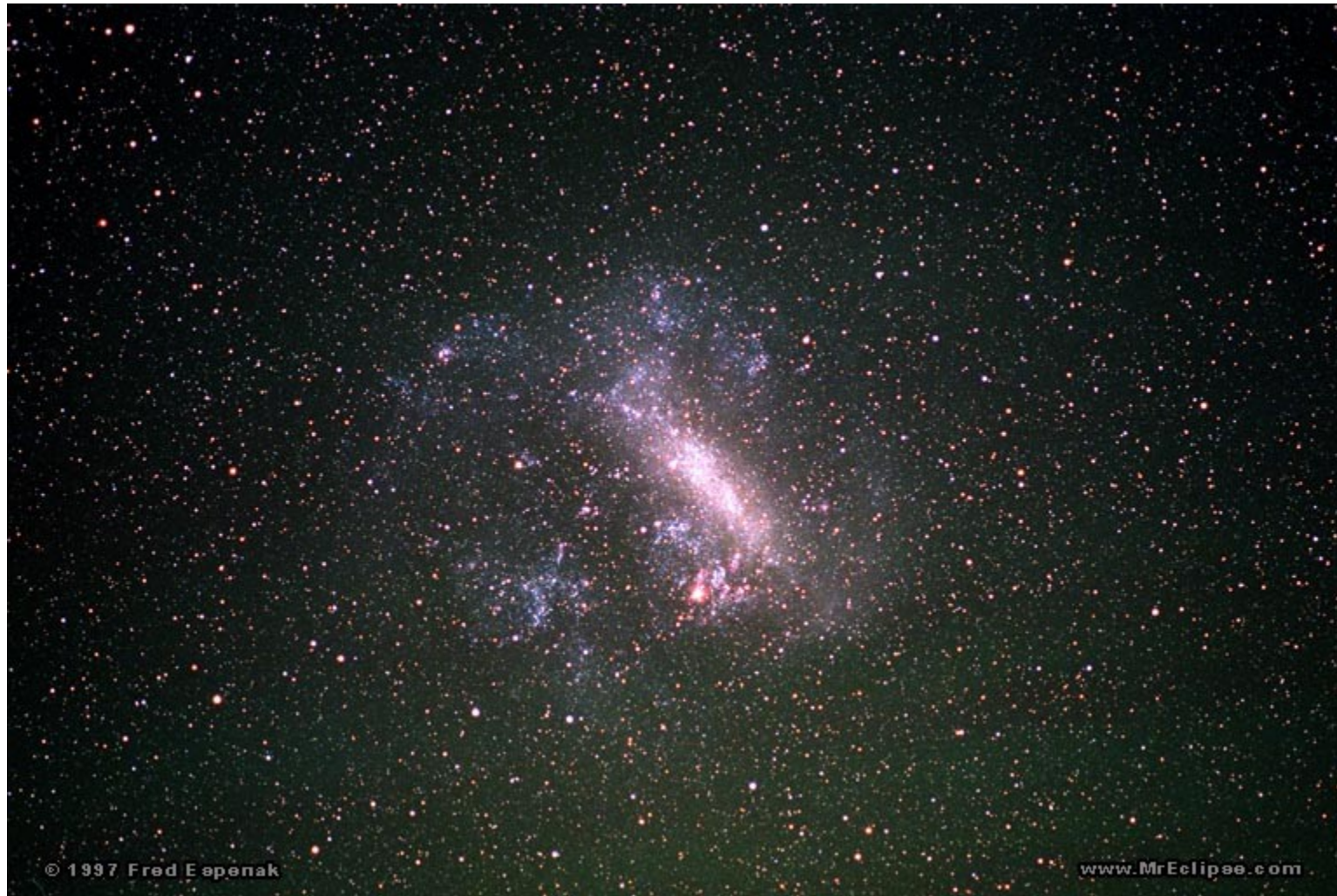
SN 1987A was a type II supernova in the Large Magellanic Cloud, a dwarf satellite galaxy of the Milky Way. ... 1987A's light reached Earth on February 23, 1987, and as the earliest supernova discovered that year, was labeled "1987A". Its brightness peaked in May, with an apparent

Inside the large Magellan cloud = the Tarantula
Nebula. The closest stellar nursery in our local
Group. Massive stars in formation. Next to
The nebula in 1987 a “ new star “ was seen
A supernovae type II





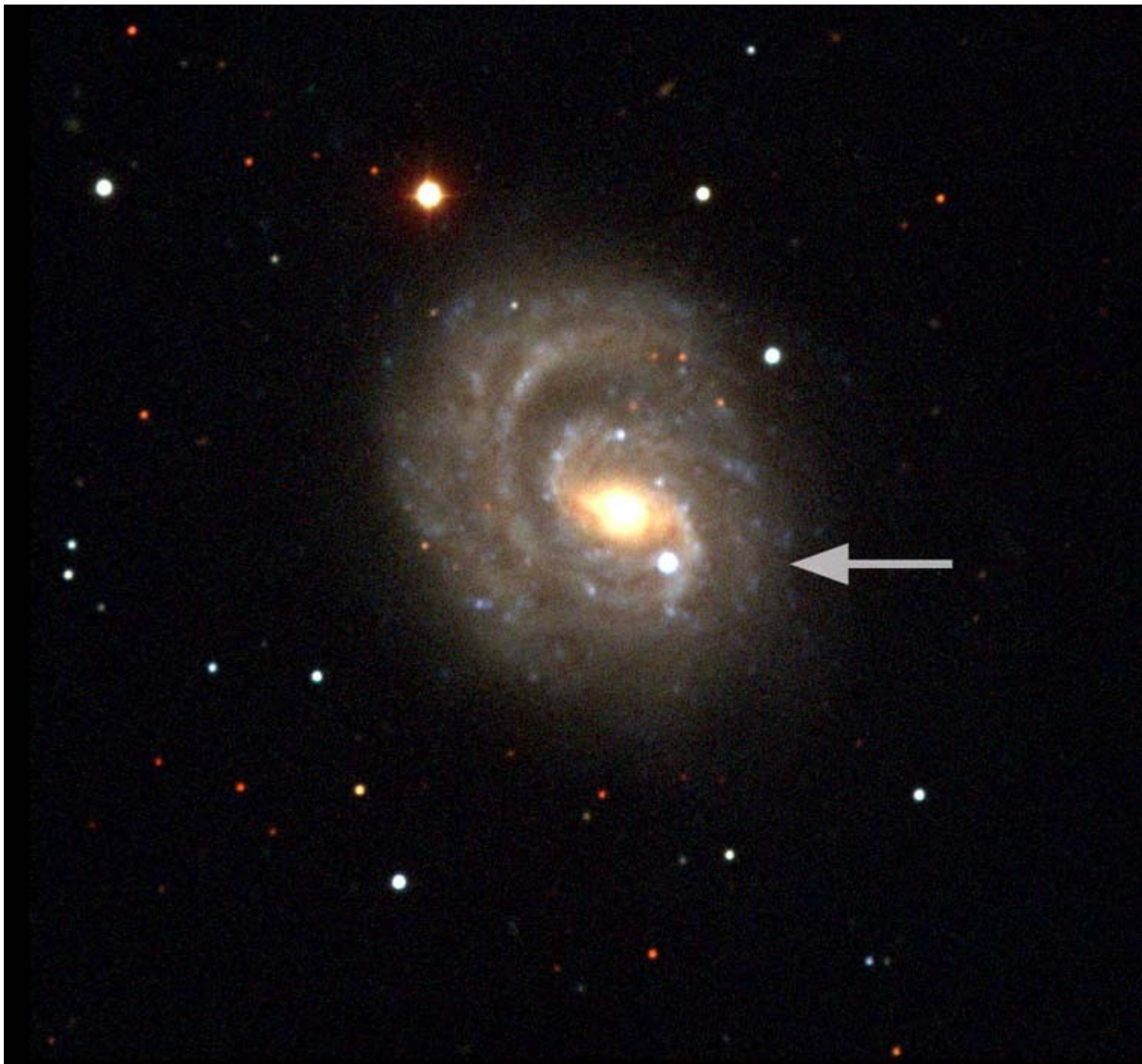
The Large Magellanic Cloud is a satellite galaxy orbiting the Milky Way, along with its companion the Small Magellanic Cloud. Visible only from the southern hemisphere, the Clouds were first brought to the attention of northern astronomers following the round-the-world expedition of Ferdinand Magellan in 1519-1522. the photo shows the small cloud and a Globular cluster. The SMC lies at a distance of 200,000 light years. The bright globular cluster nearby (above right) is named 47 Tucanae.



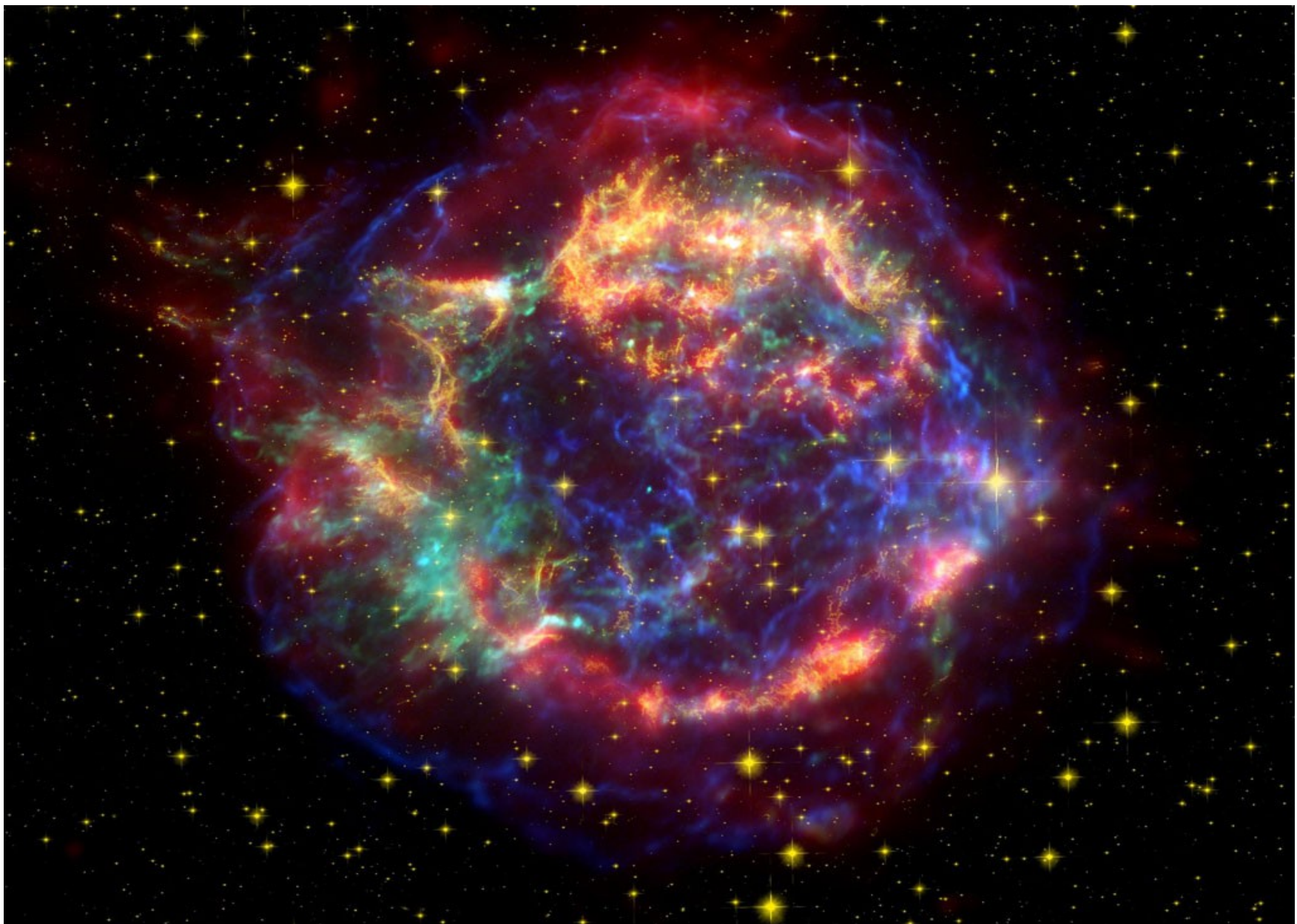
To the naked eye, the Large Magellanic Cloud appears to be a misplaced piece of the Milky Way. In reality, it is a satellite galaxy of our own. The LMC lies at a distance of 190,000 light years and is composed of approximately 25 billion stars.



Betelgeuse is a good candidate but we don't know when it will explode



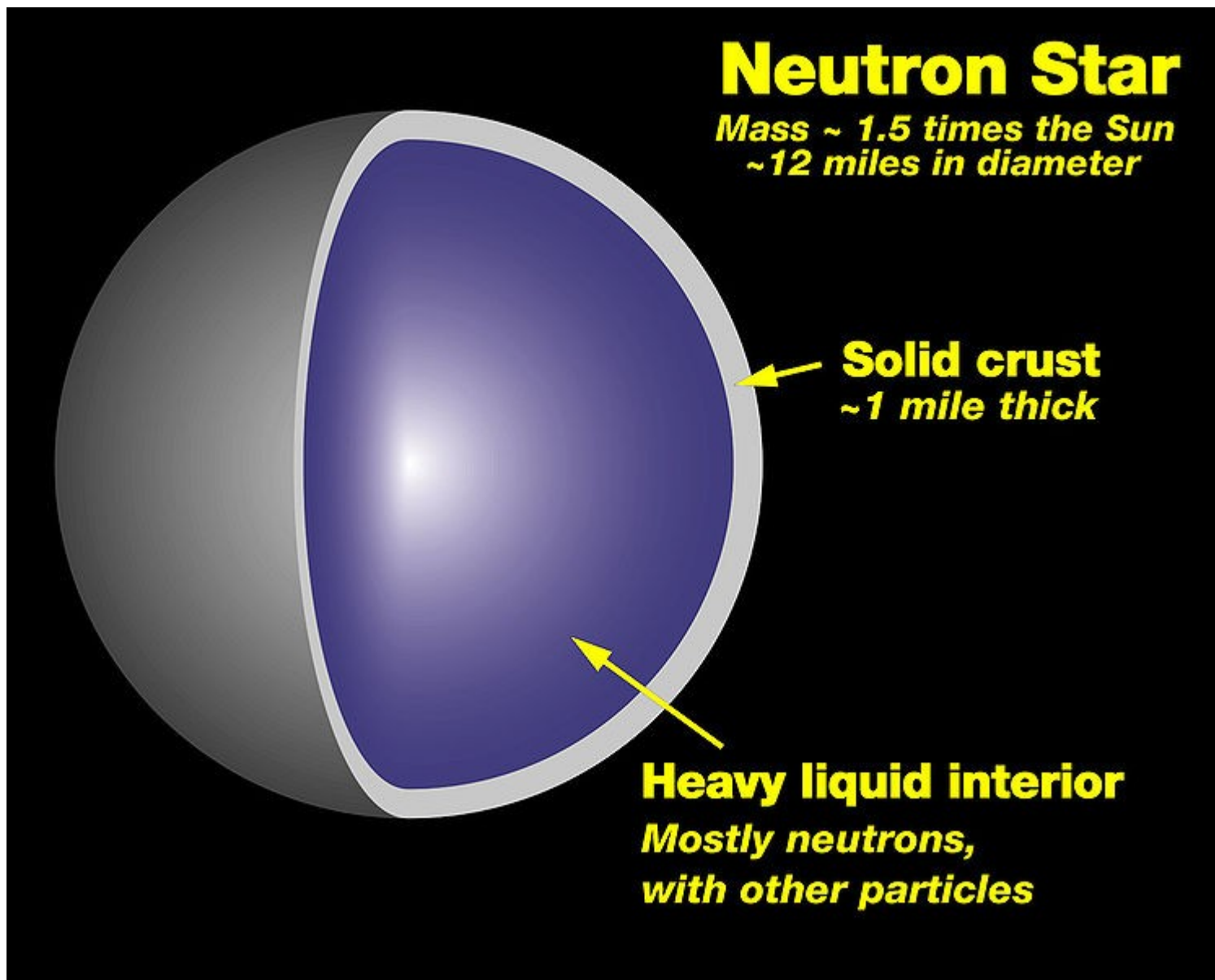
Supernova 1999em in NGC 1637 (type II)



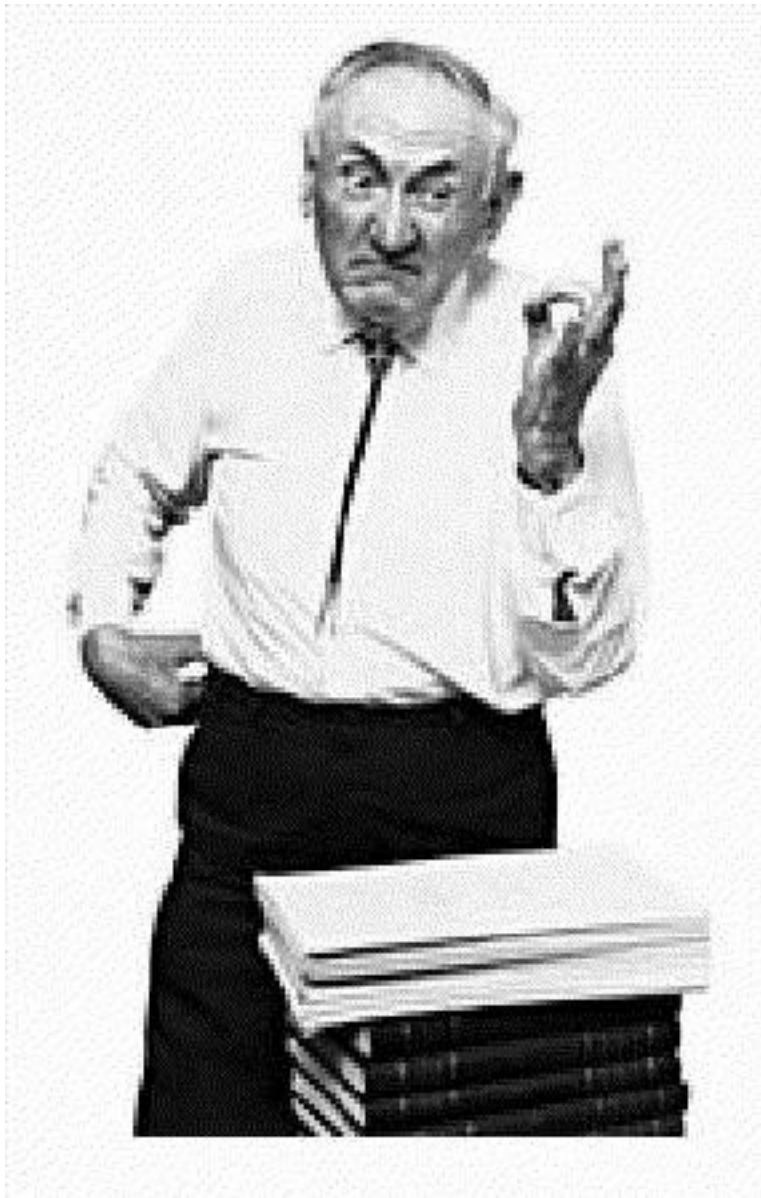
300 years old remnant of a supernova type II with a neutron star in constellation
Of Cassiopeia

Neutron Stars

- The core left over by a Type II supernova
- Held up from gravity by **neutron degeneracy pressure**
- First predicted in the 1930s, and confirmed with the discovery of pulsars in 1967 by Jocelyn Bell (her advisor got the Nobel Prize for the discovery).
- We think the maximum mass (like the Chandra limit for WDs) is $2-3M_{\text{sun}}$ and 10 km radius. This is 10^{11} kg/cm^3 !!!
- We now know about hundreds of neutron stars doing many exciting things (pulsing, surface explosions, highly magnetized).
- Neutron stars are important for testing **Einstein's General Theory of Relativity** and testing our **understanding of subatomic particles**.



Neutron stars are the left over of supernovae type II. They were predicted by Fritz Zwicky,



Source: wikipedia

In 1934 he and Baade coined the term "supernova" and hypothesized that they were the transition of **normal stars into neutron stars**, as well as the origin of cosmic rays.[5][6] It was a prescient insight that had tremendous impact in determining the size and age of the universe in subsequent decades.

In support of this hypothesis, Zwicky started hunting for supernovae, and found a total of 120 by himself (and one more, SN 1963J, in concert with Paul Wild) over a stretch of 52 years (SN 1921B through SN 1973K),[7] a record which still stands as of 2006 (the current runner-up is Jean Mueller, with 98 discoveries and 9 co-discoveries).

see also :

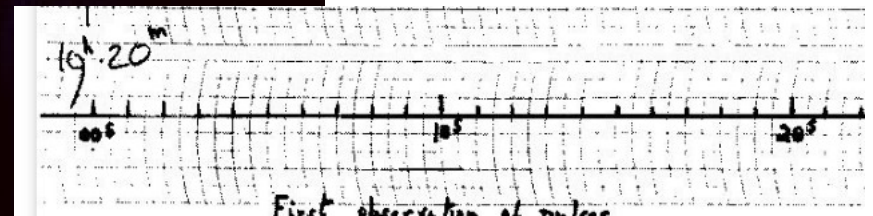
http://en.wikipedia.org/wiki/Fritz_Zwicky

**Through observations, he predicted
Dark matter**

Used to call his colleague : spherical bastards"
Not a good way to make friends.
Amazing theorist. Fritz Zwicky

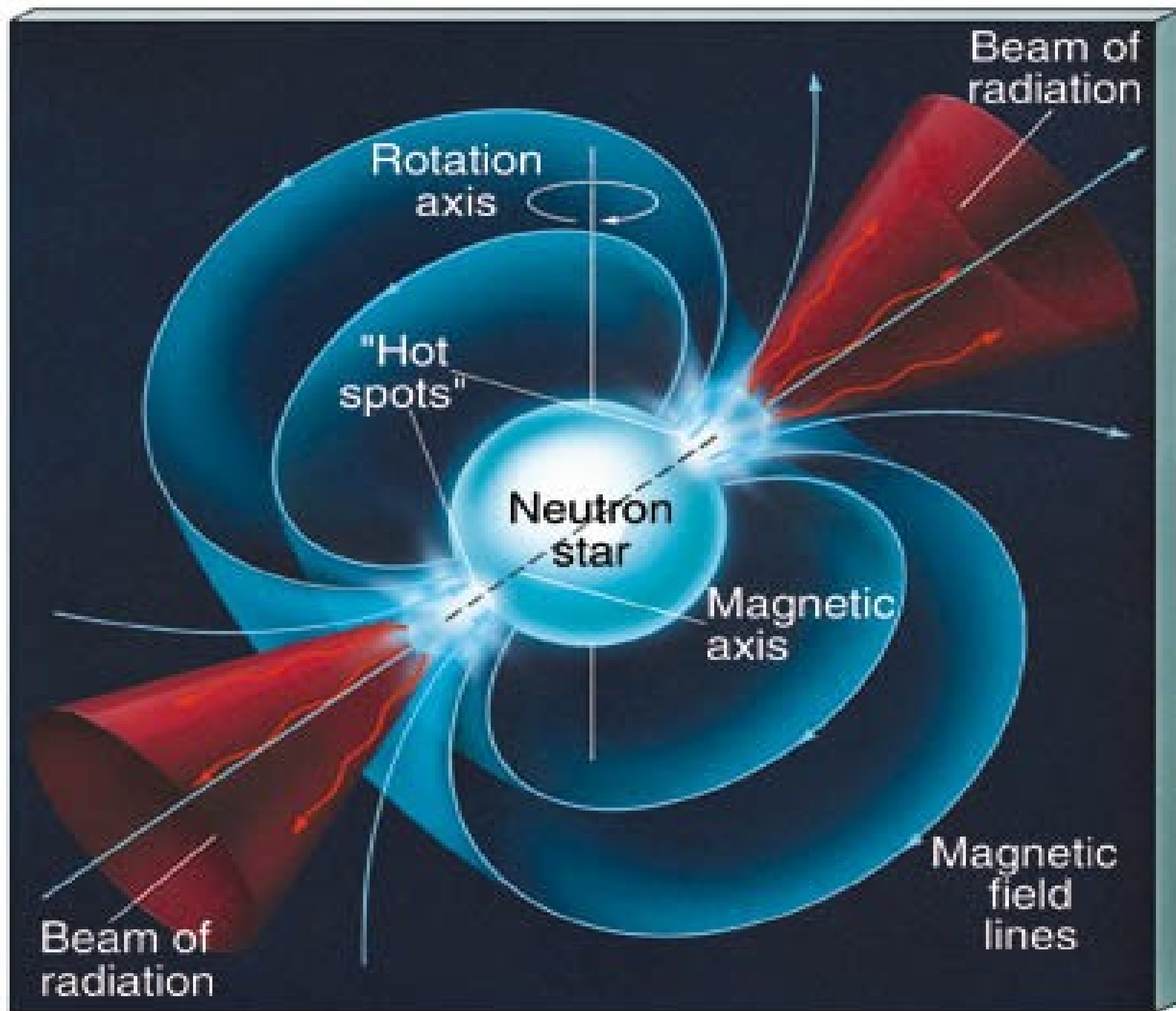


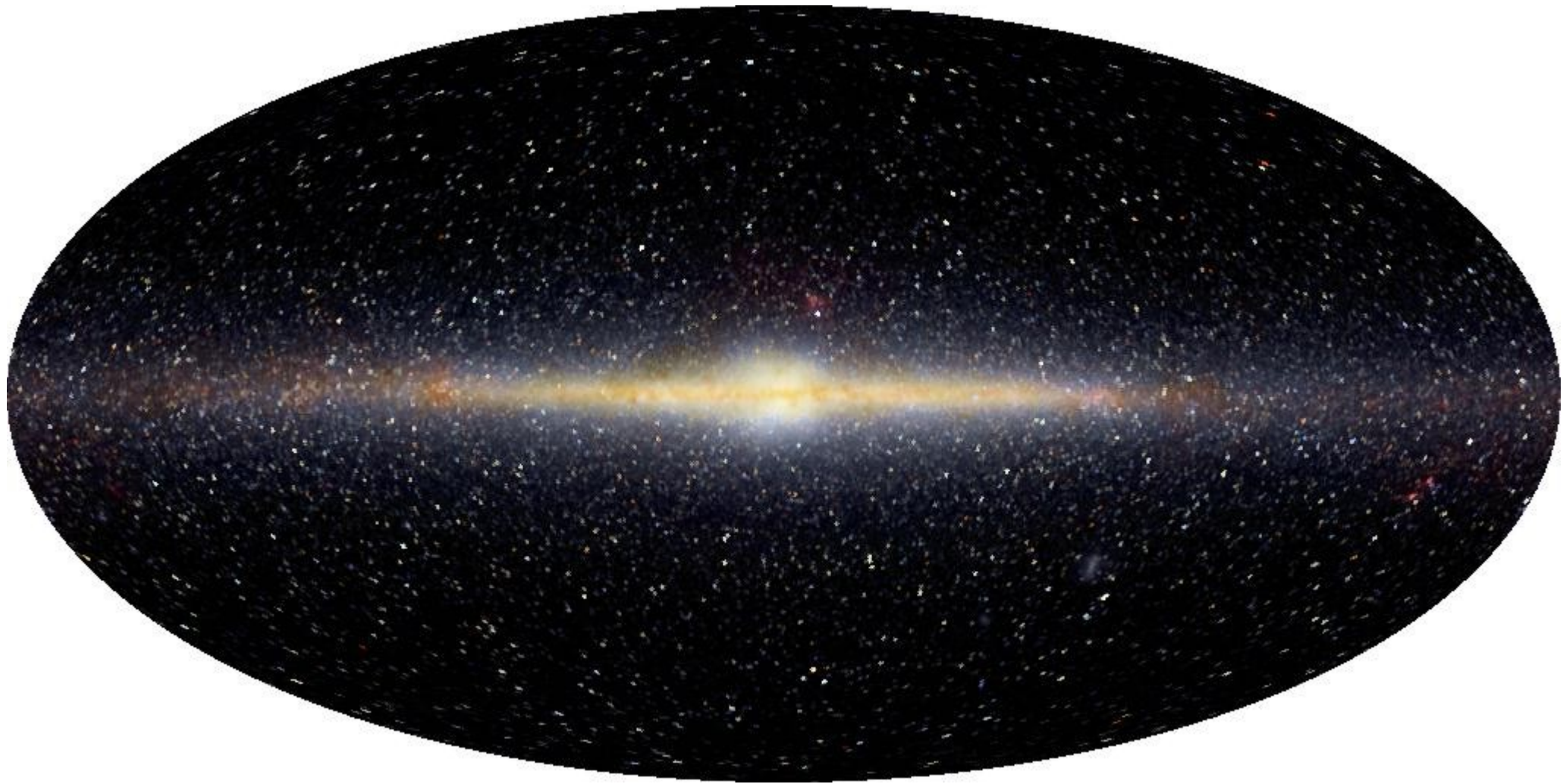
Blips recorded by
Jocelyn Bell



PSR B1919+21 is a pulsar with a period of 1.3373 seconds and a pulse width of 0.04 seconds. Discovered by Jocelyn Bell Burnell on 28 November 1967, it is the first discovered radio pulsar.

First observed by Jocelyn Bell, by accident, a graduate student at Cambridge England. She was robbed of the Nobel Prize by her advisor. Great injustice. She was using a radio telescope and observed a pulsar = highly magnetized rotating neutron star. She recorded blips in the radio wave range. See above. .





We have detected many of them in the plan of our galaxy. So they come from young Massive stars.